

Member Contributions & Manual Testing Tasks

Overall Contribution Split

Member	Role	Contribution Focus	Assigned %
Member A (Nguyễn Đình Minh Huy)	Engine & Infrastructure	Game Loop, Physics Engine, Collision Systems, Level/World Systems, Serialization, Level Editor, Replay/Rewind, Multiplayer AI (Shadow Mario, CPU opponent), Advanced Systems.	50%
Member B (Trần Gia Huy)	Entities & Gameplay	Player States & Power-ups, Enemy AI & Bosses, Items & Blocks, Game Flow/UI, Audio, Controls, Visual Effects, Accessibility, Polish & Game Feel.	50%

Manual Testing Task Division

Every one of the 127 features in `features_list.md` is assigned below to the member whose domain it falls into, grouped by that document's own 26 categories so a tester can work through it category by category with the feature numbers open side by side. This replaces the previous six-bullet-per-member sketch with a checklist that actually covers the full list.

Rows 1–14 cover features 1–97. Rows 15–25 were added when the features list was extended to 127 and cover features 98–127. Every feature number falls in exactly one row, and the ranges quoted in the two tables below cover 1–127 with no gap. The single deliberate repeat is the unnumbered accessibility row, which re-tests #95 from the other member's side and says so.

Several of the new rows need **Options > Debug Mode** switched on before the surface they test exists at all — the dev panel, the cheats panel and their `F2 – F11` shortcuts are all gated behind it. Switch it back off before any run whose score is meant to count: turning on a cheat taints the run, and a tainted run deliberately writes no high score and unlocks nothing (**#107**), so a tester who leaves it on will read that as a save/achievement bug.

Member A — Engine, Physics, World & Systems Testing

#	Category (features)	Manual test
1	Core Engine & Architecture (1–10)	Play for 10+ minutes across at least two levels; confirm no frame-rate stutter, no dangling ImGui window, and that rebinding a key in Options survives a restart. Open the debug console (<code>`</code>) and run every one of <code>help/give/lives/tp/god/spawn/difficulty/level/progress/replay/clear</code> at least once.

#	Category (features)	Manual test
2	Physics & Collision (11–16)	Deliberately test edge cases: jump into a wall corner, walk onto ice (1-2) and a conveyor (1-3) back to back, and stomp an enemy at the exact frame you also touch a block from below. Confirm no jitter or tunnelling.
3	Advanced Movement (17–22)	Wall-jump off a shaft, ground-pound from a Mega state and a normal state, crouch-slide under a 1-tile gap, and chain 3+ stomps without touching ground to confirm the combo counter increments and resets correctly on landing-idle or damage.
4	Levels & World (60–64, 96–97)	Play all 3 main worlds + Bonus Stage to their flagpole, enter and exit all 3 sub-vaults through their pipes, confirm the World Map unlocks each node in order and shows the right star-coin pips, then open the Level Editor from the main menu's MAP EDITOR row (it is its own screen, not an F1 overlay — see #64), place an entity, save, and reload it. From Main Menu → Procedural Level → Play Endless, run right for at least 2 minutes: confirm the distance counter climbs in the world-label HUD field, at least one chunk boundary is crossed with no visible break in the terrain, and dying shows "WORLD ENDLESS - Nm" with a recorded high score. Use "Generate & Play" 5+ times across different themes/difficulties and confirm every generated level is completable start to flagpole (the solvability check should mean you never see an impossible gap).
5	Save/Load & Persistence (75–78)	Save to a slot mid-level, quit, and reload it — confirm position, lives, coins and score match. Record a run's high score and confirm it appears in Options → Records.
6	Two-Player & AI Opponents (79–82)	Start a Versus Human match and a Versus CPU match (try both a Speedrunner and a Hunter profile) and a Shadow Chase match; confirm Player 2 controls respond independently and Shadow Mario visibly replays your own prior path.
7	Advanced Systems (89–95)	Hold the rewind key mid-jump and confirm the game rewinds cleanly; save and play back a replay from the console; clear the campaign once and confirm New Game+ starts with faster enemies; run the Daily Challenge twice on the same day and confirm the level is identical both times; toggle Colorblind Mode and confirm the minimap recolors.

#	Category (features)	Manual test
15	Dynamic Lighting & Day/Night (98–100)	Enter World 1-2's Ice Cavern and confirm the level is visibly darkened with a lamp of light around the player that breathes; throw a fireball as Fire Mario and confirm the shot carries its own moving light. Stand still on the World 1-1 overworld for a full 100 seconds and confirm the scene passes from day into night and back — or, faster, open Debug > Lighting, freeze the clock and drag the phase scrub, watching the sky change. In the same panel confirm the header reports whether a GPU is behind the effect, then move the player lamp radius, breathe amount, shadow tint, fireball radius/intensity and free-camera radius sliders one at a time and confirm each changes what is on screen; press Reset and confirm the shipped look returns. On a machine with no GLSL support, confirm the game still draws normally and prints one line saying lighting is off.
16	Level Editor & Custom Levels (101–104)	From Main Menu → MAP EDITOR, cycle all six tools by their shortcuts (B , E , R , I , V , P) and confirm the cursor behaviour changes for each. Place a tile, fill a rectangle, place an entity, drag it with Select, edit a property in the Inspector, then press Ctrl+Z six times and confirm every edit reverses in order, and Ctrl+Shift+Z (and Ctrl+Y) replays them — checking the History panel names each one. Click into the level-name text box, type, and confirm Ctrl+Z there undoes the <i>typing</i> and not a brush stroke. Open the entity palette and confirm every category has buttons and that each button actually places something (this is the registry check — a button that places nothing is the exact defect #104 exists to prevent). Ctrl+Shift+S to save under a name, Escape to the menu, then Main Menu → CUSTOM LEVELS: confirm the level is listed with its name and file stem, the row count on the main menu matches, and selecting it plays it. Delete every custom level and confirm the page says "NO CUSTOM LEVELS YET / MAP EDITOR > SAVE AS" rather than showing an empty list.

#	Category (features)	Manual test
17	Capture & Debug Tooling (105–109)	With Options > Debug Mode on, open Debug > Cheats and toggle all eight switches, confirming each tooltip and each visible effect: Invincible (walk into a Goomba), Infinite Lives (die and confirm the counter holds), Freeze Timer (watch the HUD clock stop), Hide HUD (clean screen), Noclip (walk through a wall with gravity off), Infinite Fireballs (fire more than two at once), Free Camera (F9 , then pan with WASD past the edge of the level and confirm the player is not dragged back into frame and the tether does not shove them), and Immortal — specifically confirm Immortal rescues: jump into a pit and confirm the player is placed back on solid ground with a reason printed, rather than falling forever or dying. Move the time-scale dial to 0.1× and confirm the simulation crawls while the panel and the mouse stay responsive. Then confirm the tainted-run rule: with a cheat used, finish or fail the level and confirm no high score is written to Options → Records and no achievement unlocks; turn the cheat off mid-level and confirm the run is still refused (the taint persists on purpose). Confirm F2 – F11 flip the same switches as the checkboxes and print a line each. Finally open the console (`), type g and press Tab — confirm it completes to give with a trailing space; type l and Ta b — confirm it completes only as far as the candidates agree and lists them; type give then Tab and confirm nothing is completed past the space.
18	Camera & Simulation Budget (117–119)	Single player: run right at full speed on World 1-1 and confirm the player is never carried off the side of the screen, then deliberately fall into a pit and confirm they keep dropping <i>below</i> the visible view until the death fires — the tether must not arrest them at the bottom edge. Enable Free Camera and then Noclip and confirm the tether releases in both cases. Two-player Versus: let Player 1 lose all lives and confirm the camera then follows Player 2 for the rest of the match instead of stopping dead, and that the HUD shows an eliminated badge naming Player 1's character with their final coins and score frozen. Start the next level and confirm neither player is still marked eliminated. For the update gate, open the dev panel during a long Endless run and read the entity census — confirm the frozen count rises as chunks accumulate while the thought count stays bounded, and confirm nothing frozen is ever visible on screen: walk back and forth past a chunk boundary and check that enemies are already moving by the time they are drawn.

#	Category (features)	Manual test
19	Attract Mode (120)	Reach the main menu and leave the keyboard alone for 30 seconds (or set <code>SUPERMARIO_ATTRACT_IDLE_SECONDS</code> low, or press <code>F 5</code>): confirm the game starts playing itself from the bundled demo, that what is on screen is the real game being driven rather than a video, and that any key returns to the menu. Then confirm the gate: sit idle on the LOAD GAME page and on the MULTIPLAYER page and confirm the demo does not interrupt the choice being made. Temporarily rename <code>assets/replays/attract_demo.json</code> and confirm the game returns to the menu with a message rather than showing a demo that plays nothing.
20	Save Management & Endless Refinements (126–127)	On the LOAD GAME page, select a populated slot and request delete: confirm the confirmation page appears with "keep" preselected, that cancelling leaves the file intact, and that confirming empties the slot, plays the block-break cue rather than a positive chime, and re-reads the page from disk so the slot now reads Empty. Confirm requesting a delete on an empty slot or on the BACK row gives the blocked-action blip and does nothing. Check afterwards that the <i>other</i> two slots are untouched. Then run Endless Mode for long enough to pass several chunk boundaries: confirm the HUD clock never counts down and never kills the player, and confirm bosses arrive at a fixed interval as milestones — not one Bowser per chunk once the difficulty ramp turns Hard.

Member B — Entities, Gameplay & Polish Testing

#	Category (features)	Manual test
1	Characters & Player States (23–31)	Play a full level as each of Mario, Luigi, Toad and Peach, noticing the speed/jump differences. Take every power-up in sequence (Small → Super → Fire → Cape, and separately Mini and Mega) and confirm the hitbox and sprite change each time; take a Star while in Fire form and confirm you return to Fire form when it ends.
8	Enemies & AI (32–41)	Encounter every enemy type at least once (place the editor-only ones — Paratroopa, Boo, Thwomp, Lakitu, Bullet Bill, Chain Chomp — via the Level Editor if they are not in your current playthrough) and confirm each one's movement strategy matches its description (patrol, chase, fly, timer emergence, tethered lunge).
9	Bosses (42–46)	Fight Boom Boom in 1-2 and Bowser in 1-3 to completion. Confirm the boss health bar only appears during the fight, the arena will not let you leave or reach the flagpole until the boss is dead, and each boss's phase 2/3 attack pattern actually changes.

#	Category (features)	Manual test
10	Items & Blocks (47–59)	Trigger every item at least once (all 7 power-ups, a POW Block, a P-Switch, a Trampoline) and every block type (Question, Brick, a Moving Platform, a Falling Platform, ice and conveyor tiles, a Warp Pipe). Finish a level and confirm the castle's flag rises and you visibly walk up to its door before the summary screen appears.
11	Game Flow & UI (65–71)	Walk the full Main Menu → Pause → Options → Records loop. Toggle the Minimap with M and again with Tab (both keys work — #68) mid-level, open the Statistics screen, and die on purpose to confirm the death tally and instant "Retry Level" both work.
12	Audio (72–74)	With sound on, confirm each listed SFX actually plays for its trigger (jump, coin, stomp, power-up, damage, fireball, death, flagpole, pipe, ground pound, P-Switch, achievement) and that the music changes between the menu, world map, a level, a boss fight and Game Over. Move both volume sliders and confirm the change is audible.
13	Controls (83–84)	Rebind at least 3 actions to new keys in Options, confirm they work in-level, then use "Reset Controls" and confirm the defaults return.
14	Visual Effects (85–88)	Break a brick, collect a coin, stomp an enemy with a combo active, and take damage — confirm a particle burst, the correct death animation, and the invincibility flash all appear as described.
21	Warp Pipe Geometry (110–111)	Take the top-entry pipe into a sub-vault by standing on the rim and pressing Down, then leave the vault through its side-entry exit pipe by <i>walking into the mouth on its face</i> — confirm the exit pipe is drawn as an L-bend whose shaft leaves the top of the frame, so it reads as "goes up", and confirm it can be entered as Small, Super and Fire Mario alike (the trigger measures feet, so every form must enter the same mouth). Confirm the entry is animated: the player is walked into the mouth over a moment while the screen fades, and nothing pushes them back out part-way. Then use Main Menu → PROCEDURAL LEVEL → Generate & Play until you reach a generated vault and confirm its exit pipe is a side-entry L-bend too, not a top-entry stub.

#	Category (features)	Manual test
22	Boss Fight Structure (112–114)	In World 1-3 as Fire Mario, land four fireballs on Bowser and confirm he staggers for about three seconds, that the HUD counts down the fireballs still owed, and that a staggered Bowser can be stomped without the usual invulnerability frames. Then take the non-combat route: on Normal difficulty confirm two axes must be reached and that only the last one drops the bridge; repeat on Easy (one axe) and Hard (three axes), confirming that Hard is <i>harder</i> because the bridge must be crossed three times under fire. When the bridge goes, confirm Bowser falls into the lava and burns down there rather than passing through the lava tiles or falling forever, and that no "boss left the level" warning appears. For procedural arenas, use Generate & Play on Hard difficulty several times and confirm each generated level contains a boss arena placed before the flagpole, never a half-built one.
23	Enemy Behaviour Detail (115–116)	Stand under the Lakitu in World 1-1 and let it drop Spinies: confirm each one arrives as an egg with its own sprite, is tossed sideways, falls, and hatches into a walking Spiny on landing — not already walking in mid-air. Then stay put until three Spinies are alive at once and confirm the Lakitu drops a Fire Flower instead of a fourth Spiny, and that the drop timer does not bank up and fire a volley once you move away. Also confirm the Spinies placed directly in Worlds 1-2 and 1-3 are walking normally (they spawn flush against the ground, so they hatch immediately).
24	Presentation & Audio Detail (121–123)	With sound on, walk on World 1-1 grass, then on World 1-2 ice, then on a metal/other solid surface, and confirm the footstep cue changes between the three and stops when jumping, crouching or wall-sliding; in two-player, confirm both players' footsteps run on their own cadence instead of merging into one sound, and that footsteps sit below the gameplay cues in the mix. In the menus, move the highlight one row at a time and confirm every move plays a blip, that a blocked move (a locked character, the end of a list, confirming an empty slot) plays the same short cue, and that deleting a save plays the destructive break cue instead. Then chain 5+ stomps without landing and confirm the combo cue rises in pitch with each hit and stops rising once the chain is long.
25	Visual Effects, Continued (124–125)	Take a Star, then stomp an enemy while it is active: confirm the enemy spins away with a sparkle burst rather than the ordinary upside-down flip, and confirm a fireball kill in the same run still uses the flip, so the two are visibly distinguishable. Then wall-slide down a shaft and confirm dust comes off the wall, and stand in the water of a sub-vault and in the lava band of World 1-3 and confirm ambient bubbles and lava embers respectively — three emitters that fire from where the player is rather than from a gameplay event.

#	Category (features)	Manual test
—	Accessibility (95 , cross-referenced with Member A's #7)	Confirm the colourblind palette swap is visible on the minimap and any debug outlays without needing the debug console.

Cross-cutting: play at least one full session together

However the two domains above are split, finish with one full co-op or versus session played together end to end (Main Menu → a full level → Victory → World Map → a second level → Game Over or campaign clear), so at least one manual pass exercises the transitions between every system in this table rather than each system in isolation.